Proposition 8: no linear bound for the local edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Proposition8_LocalEditLowerBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Proposition 8 (Section 3.2, p. 7, l. 262–263; construction and proof in Appendix D.1, pp. 25–28, l. 791–863): *For every even $r \geq 2$, there are graphs G_r, H_r on $\Theta(r^3)$ vertices such that $d_1(G_r, H_r) = r$ and $d_1^{\mathrm{OT}}(G_r, H_r) = 1$.* Normalised, this is Theorem 3 (2) (p. 4, l. 125): $\delta_1(G_r, H_r) = r \cdot \delta_1^{\mathrm{OT}}(G_r, H_r)$. Two remarks in the text concern the same graphs: after Proposition 8 (p. 9, l. 295) "Proposition 8 rules out any bound of the form $\delta_1 \leq c \cdot \delta_1^{\mathrm{OT}}$", and after Theorem 9 (p. 9, l. 318–320) "there is no uniform bound of the form $\delta_1 \leq C \cdot (\delta_1^{\mathrm{OT}})^{\alpha}$ with $\alpha > 2/3$".

Lean. With G_r, H_r the concrete graphs `GsimOT.Prop8.Gr`, `GsimOT.Prop8.Hr` on the vertex type `Vr r` (core $\sqcup$ markers $\sqcup$ clique), and for every $r \in \mathbb{N}$ with $2 \leq r$ **and** r **even**:

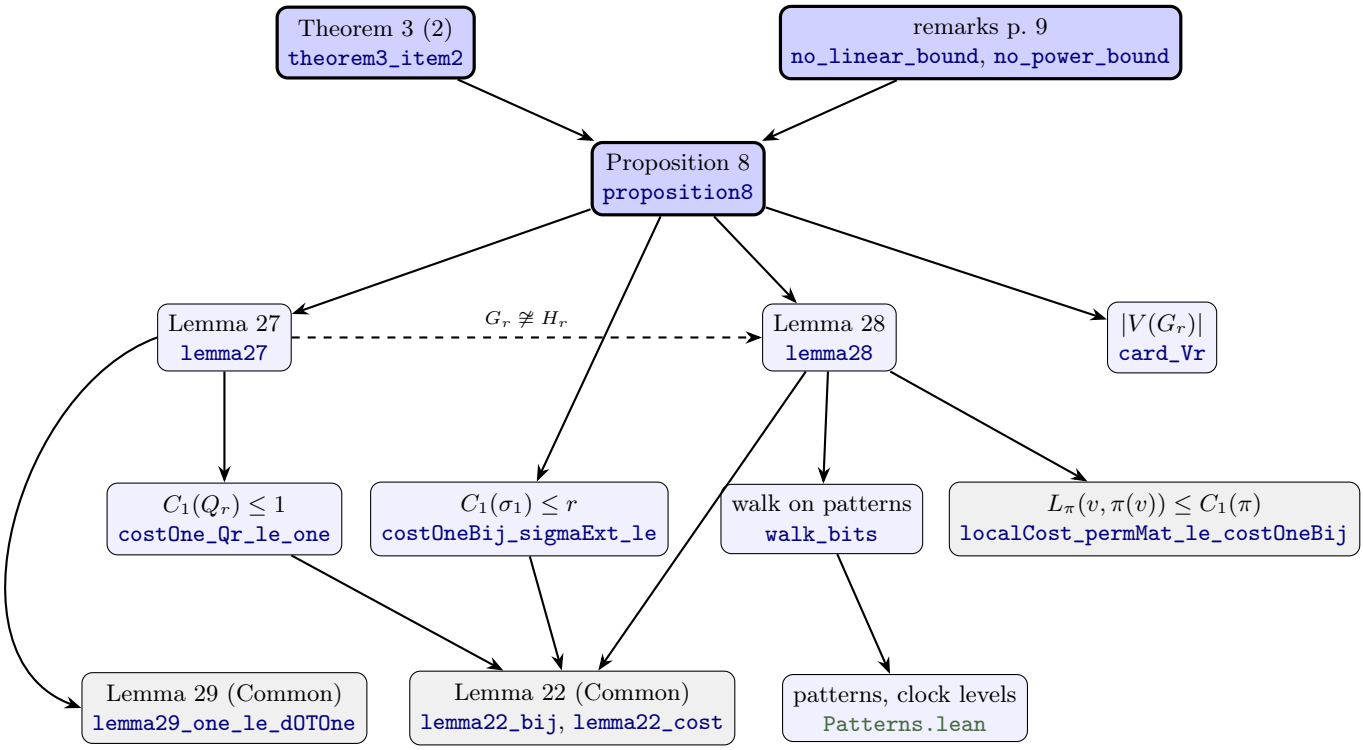
- `GsimOT.Prop8.proposition8`: $d_1(G_r, H_r) = r$, $d_1^{\mathrm{OT}}(G_r, H_r) = 1$ and $r^3 \leq |V(G_r)| \leq 19r^3$; moreover `card_Vr`: $|V(G_r)| = 5r^3 + 10r^2 - r + 4$.
- `GsimOT.Prop8.theorem3_item2` (Theorem 3 (2)): $\delta_1(G_r, H_r) = r \cdot \delta_1^{\mathrm{OT}}(G_r, H_r)$.
- `GsimOT.Prop8.no_linear_bound`: for every $c \in \mathbb{R}$ some even $r \geq 2$ has $\delta_1(G_r, H_r) > c\,\delta_1^{\mathrm{OT}}(G_r, H_r)$.
- `GsimOT.Prop8.no_power_bound`: for all $C \in \mathbb{R}$ and $\alpha > 2/3$ some even $r \geq 2$ has $\delta_1(G_r, H_r) > C\,(\delta_1^{\mathrm{OT}}(G_r, H_r))^{\alpha}$.

The Lean statements are exactly the paper's (with explicit constants for $\Theta(r^3)$).

2 Where the proof lives

File	Content
`Construction.lean`	ℓ, n_X, n_M, core X and markers $M = C \cup T \cup B$; the adjacency table as patterns `pat`; G_r^0, H_r^0 (`G0`, `H0`); degrees (`deg_bip_le`, `deg_H0_marker`). Appendix D.1, l. 792–809.
`Patterns.lean`	Consecutive patterns differ in one coordinate (l. 813–818, Fig. 6); how often each marker changes (`nchg_*`); clock levels (`clockLevel`) and their properties used in Lemma 28.
`LockedGraphs.lean`	Lemma 22 applied with markers sorted by degree, $\Delta = n_X$, $\tau = r$, $\delta = 1$: `Gr`, `Hr`, the hypotheses `lock_hyp_*`, $|V(G_r)|$ (`card_Vr`); the shifts σ_i (`sigma`).
`Lemma27_FractionalCost.lean`	Lemma 27, upper bound: Q_r (`Qr`, `QrExt`) and $C_1(Q_r) \leq 1$ (`costOne_Qr_le_one`); also $C_1(\sigma_i) \leq r$ (`costOneBij_sigmaExt_le`).
`Lemma28_BijectionCost.lean`	Lemma 28: the walk argument (`walk_bits`, `lemma28`).
`Proposition8.lean`	Lemma 27 (= 1, `lemma27`), Lemma 28 as stated (`lemma28_dOne`), Proposition 8, Theorem 3 (2), the two remarks.
`Common/Observation21_DegreeBound.lean`, `Common/Lemma22_LockingMarkers.lean`, `Common/Lemma29_Integrality.lean`	Observation 21, Lemma 22, Lemma 29 (shared).

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. No cited result is used. The dashed arrow is the derivation of $G_r \ncong H_r$ from Lemma 28 (see Section 6). Lemma 27 and the bound on $C_1(\sigma_1)$ also use the pattern facts of `Patterns.lean`. Observation 21 is used only inside the proofs of Lemmas 22 and 29 in `GsimOT/Common` (see Section 6).

4 The proof step by step

Construction (Appendix D.1, pp. 25–26, l. 792–809). $\ell = r^2 + 2r$ (`ell`), $X = \{u\} \cup \{v_{i,k}\}$ (`Core`), markers $c_{j,h}$, $t_{i,h}$, b_i (`Marker`; indices 0-based in Lean). The table of l. 800 is the definition of `pat r i t`, the pattern of branch i at time t; G_r^0, H_r^0 are `G0`, `H0`, and `deg_H0_marker` is $\deg_{H_r^0}(z) = \deg_{G_r^0}(z) + \mathbf{1}[z \in C]$ (l. 802). Lemma 22 is applied with the markers sorted non-decreasingly by degree (`markerEnum`, `markerEnum_mono`), $\Delta := n_X$, $\tau := r$, $\delta := 1$ (l. 803–809): `Gr`, `Hr`; the hypotheses of Lemma 22 are `lock_hyp_deg_G`, `lock_hyp_deg_H`, `lock_hyp_delta`. In Lean, G_r^0 and H_r^0 are defined on the same type, so no relabelling (l. 806–807) is needed.

Patterns (l. 810–818, Fig. 6). $\chi_{G_r}(u) = \mathtt{pat}\, i\, 0 = 0$ (`pat_zero`), $\chi_{H_r}(u) = \mathtt{pat}\, i\, \ell = \mathbf{1}_C$ for even r (`pat_ell`); consecutive patterns differ in at most one coordinate (`ham_pat_succ_le_one`, via `change_clock_iff`, `change_tag_iff`, `change_bit_iff`).

Lemma 27 (p. 27, l. 820–837). The shifts σ_i are `sigma`/`sigmaExt` and $Q_r = \frac{1}{r}\sum_i P_{\sigma_i}$ is `Qr`/`QrExt`. Since Q_r fixes markers and K, the second item of Lemma 22 (`lemma22_cost`) moves the computation to (G_r^0, H_r^0). Supported core pairs cost their Hamming distance ≤ 1 (`localCost_sigma_core`, `ham_shift_le_one`); a marker costs the Q_r-mass of the changes at it (`localCost_sigma_marker`), bounded by 1, $2/r$, 1 for clocks, tags, bits (`nchg_clock_le`, `nchg_tag_le`, `nchg_bit_le`, `sum_nchg_le`). Hence $C_1(Q_r) \le 1$ (`costOne_Qr_base_le_one`, `costOne_Qr_le_one`). The equality $d_1^{\mathrm{OT}} = 1$ (`lemma27`) uses Lemma 29 (`lemma29_one_le_dOTOne`) with $G_r \ncong H_r$, which is derived from Lemma 28 ($d_1 \ge r > 0$).

Lemma 28 (p. 27, l. 838–859). In `lemma28`: if $C_1(\pi) < r$, Lemma 22 (1) (`lemma22_bij` with $\tau = r$) shows that π fixes all markers and permutes X; this gives the core permutation P. The local cost at a core vertex is at least the Hamming distance of $\chi_{G_r}(x)$ and $\chi_{H_r}(\pi(x))$, hence $< r$ (`sum_Vr_ge_marker`). The walk $u, P(u), \dots, P^s(u) = u$ (s the least period) gives the pattern sequence $x_0, \dots, x_s$ (l. 842–844). The rest is the purely combinatorial `walk_bits`, which follows l. 844–857 line by line: clock levels (`clockLevel`)

of consecutive patterns differ by $< r$ (`clockLevel_le_add_ham`); β is the first index with full level, $\alpha < \beta$ the last with level 0; all middle patterns lie on one branch i because patterns of different branches with middle clock level differ in $\geq r$ tags (`pat_tag_of_middle`, `ham_ge_of_middle_ne`); ι_j is the first index in $(\alpha, \beta]$ with level $\geq jr$, $c(x_{\iota_j - 1}) \in ((j-1)r, jr)$ forces block j and bit value $j \bmod 2$ (`div_eq_of_clockLevel`, `pat_bit_self`); counting changes (`card_changes_step`) gives $\geq r$ alternations of b_i. Back in `lemma28`, the r alternations are r distinct core vertices x with $A_{G_r}(b_i, x) \neq A_{H_r}(b_i, \pi(x))$, so $L_\pi(b_i, b_i) \geq r$, a contradiction.

Proof of Proposition 8 (pp. 27–28, l. 860–863). $d_1^{\mathrm{OT}} = 1$ is Lemma 27. For $d_1 \leq r$, σ_1 fixes markers and K, so `lemma22_cost` again reduces to (G_r^0, H_r^0) and `costOne_sigma_base_le` shows that *every* local cost of σ_1 is $\leq r$ (`costOneBij_sigmaExt_le`); with Lemma 28, `dOne_Gr_Hr`: $d_1 = r$. The order is `card_Vr`, `card_Vr_bounds`.

Theorem 3 (2) and the remarks (p. 4, l. 125; p. 9, l. 295 and l. 318–320). `theorem3_item2` divides by n. `no_linear_bound` takes $r > c$. `no_power_bound`: $\delta_1 = r/n$, $\delta_1^{\mathrm{OT}} = 1/n$ and $n \leq 19r^3$ give $n^{1-\alpha} \leq 19\, r^{3(1-\alpha)}$, so $\delta_1 \leq C(\delta_1^{\mathrm{OT}})^\alpha$ would imply $r^{3\alpha-2} \leq 19C$, false for large r.

5 Assumptions and trusted base

No cited result and no hypothesis. All results depend only on `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`).

6 Notes on the formalisation

- **$G_r \not\cong H_r$.** The proof of Lemma 27 obtains $d_1^{\mathrm{OT}}(G_r, H_r) \geq 1$ from Lemma 29 since $G_r \not\cong H_r$ (p. 27, l. 837). The Lean derives $G_r \not\cong H_r$ from Lemma 28: every bijection has cost $\geq r > 0$, while an isomorphism has cost 0. It does not use the argument of the proof idea (p. 8, l. 282–283), which obtains the equality from Observation 21 since u is isolated in G_r but H_r has no isolated vertex.
- Lemma 27: the paper states the local costs at clock markers, tags and bits (1, $2/r$, 1; l. 836–837); the Lean proves these values as upper bounds, which give $C_1(Q_r) \leq 1$.
- Lemma 28: at a core vertex x the Lean uses that $L_\pi(x, \pi(x))$ is at least the Hamming distance of the patterns (the paper: equal to it, l. 840–841). For patterns on different branches only Hamming distance $\geq r$ is used (the paper notes $\geq 2r$, l. 852).
- The order $\Theta(r^3)$ is stated with explicit constants, $r^3 \leq |V(G_r)| \leq 19r^3$ (`card_Vr_bounds`), derived from the exact count `card_Vr`.
- The asymptotics $\delta_1^{\mathrm{OT}} = \Theta(r^{-3})$, $\delta_1 = \Theta(r^{-2})$ (p. 9, l. 318–319) are not stated separately; their consequence (no exponent $\alpha > 2/3$) is `no_power_bound`.

7 How to check this result

- Build: `lake build GsimOT.Proposition8_LocalEditLowerBound.Proposition8` (or `bash build.sh` for the whole library, which also runs `scripts/CheckAxioms.lean`).
- Axioms: `#print axioms GsimOT.Prop8.proposition8` (and `theorem3_item2`, `no_linear_bound`, `no_power_bound`); expected: `propext`, `Classical.choice`, `Quot.sound`.
- `Proposition8_correspondence.pdf` in this folder lists every paper item with its Lean declaration.